

# **MSE 380: Microstructure of Organic Materials**

## **Module 3: Mechanical Properties of Polymers**

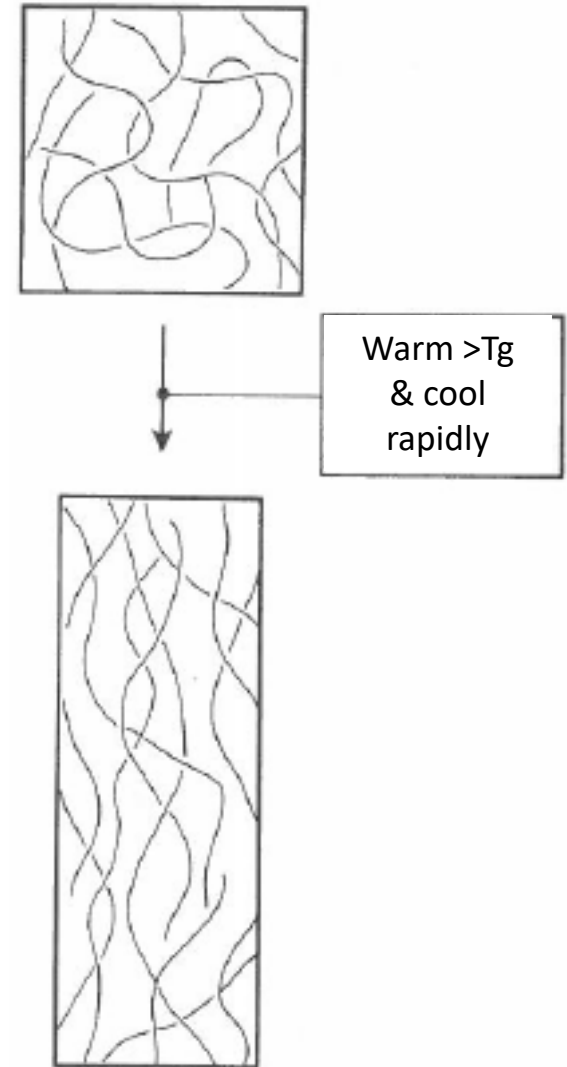
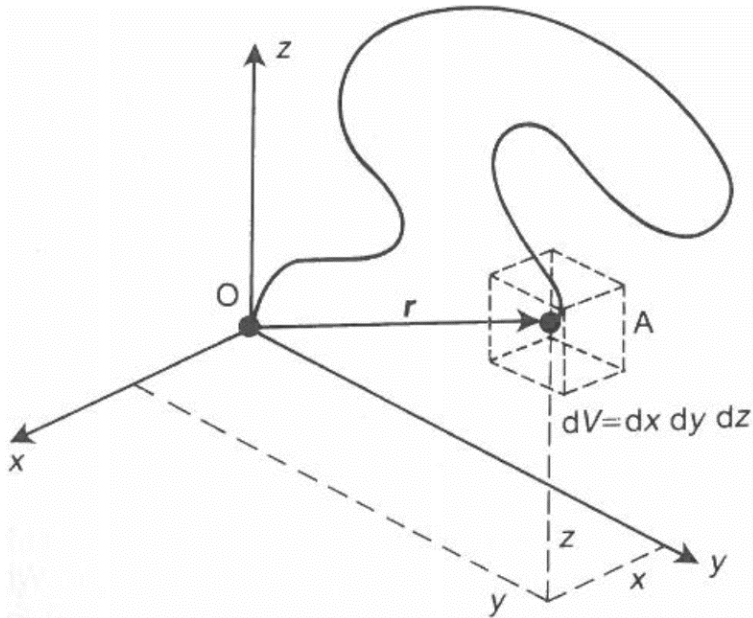
# Objectives:

By the end of the module, we should:

- i) Explain stress-strain responses of different polymeric materials including block polymers (solids)
- ii) Be able to demonstrate why chain or fiber alignment induces anisotropy in mechanical behavior of a material
- iii) Estimate the chain-direction Young's modulus of a polymer based on its crystal structure and IR/Raman spectra
- iv) Extend our approaches to modelling complex polymer systems
- v) Understand how  $E$  changes with  $T$  over multiple polymer classes
- vi) Relate elasticity to thermodynamics

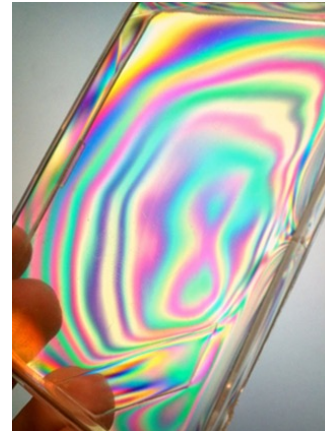
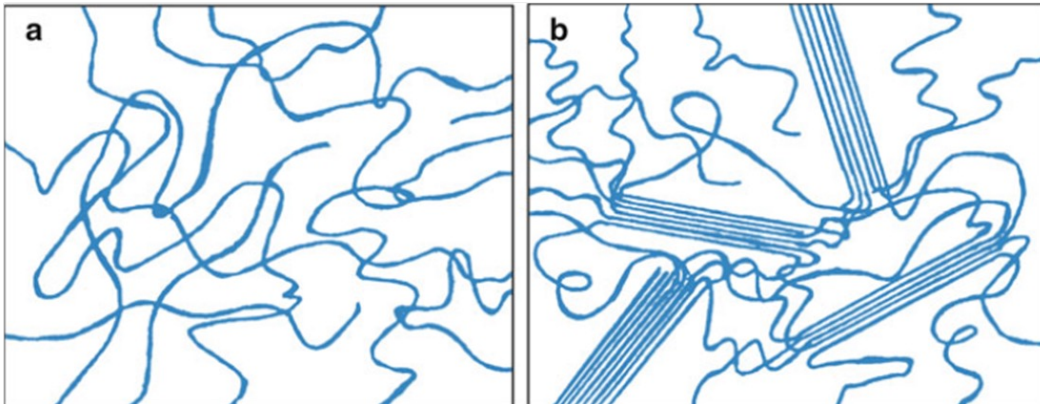
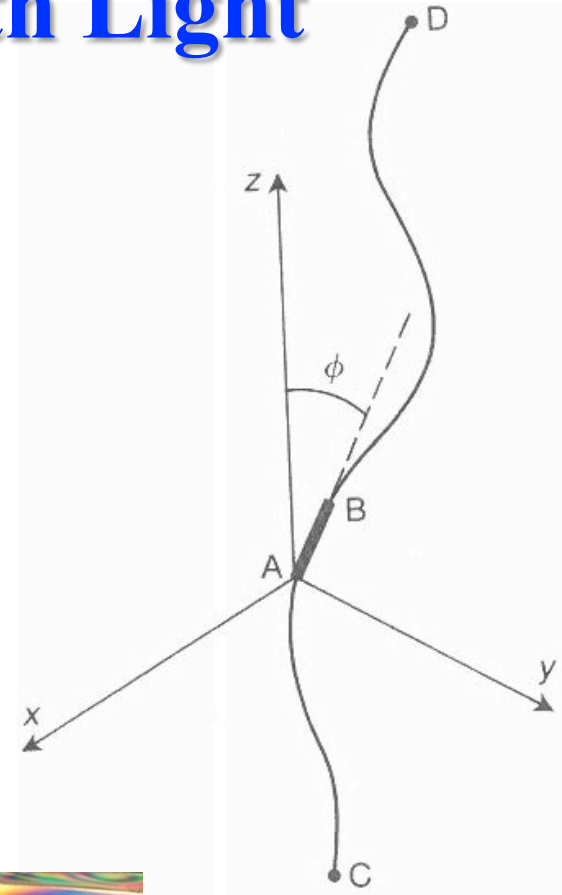
# Start with the Polymer Chains

- We observed that the properties of the polymer depend on its thermal/processing history
- Why is this important (who cares)?
- Induces anisotropy in polymeric materials
- How does this affect stress-strain responses?



# Example: Interaction with Light

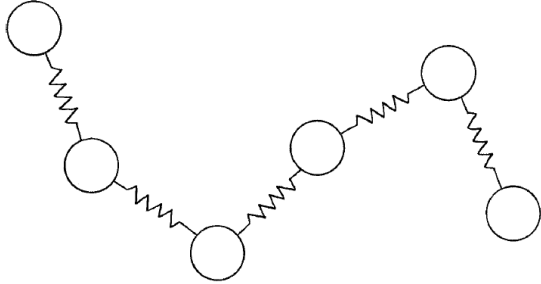
- Suppose polymer chains are aligned along the Z-axis, what happens when plane polarized light is passed through the polymer in the y-direction?
- Refractive indices perpendicular,  $n_{\perp}$ , or parallel,  $n_{\parallel}$ , are different.
- For most polymers  $n_{\parallel} > n_{\perp}$ . The difference is birefringence ( $\Delta n$ ).  
$$\Delta n = n_{\parallel} - n_{\perp}$$
- **Orientation factor,  $f$ ,** and **birefringence,  $\Delta n$ ,** are related by;  
$$\Delta n = \Delta n^o f$$
- $\Delta n^o$  is the birefringence when all molecules are perfectly aligned and is specific to any particular chemical structure



# Recall: Polymers reptate in the solid state

## 1. Rouse-Bueche Theory:

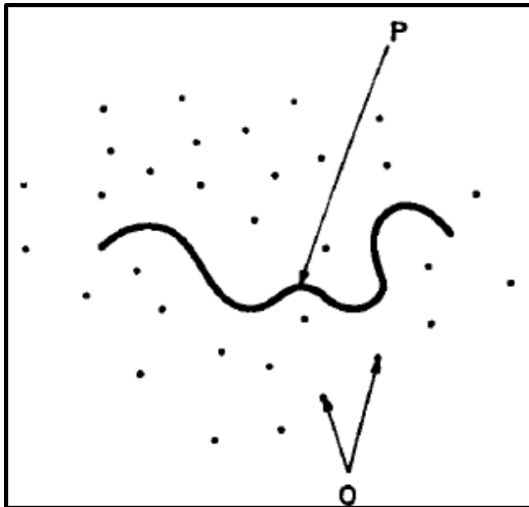
- We have the beads, so;



- Adopting this model, can we use Hooke's law to understand the mechanical properties of polymer?

## 2. De Gennes Theory:

- Propagation of defect currents induce reptation (movement in the solid state)



- This would imply that the polymers 'flow'
- Should we then consider them Newtonian?

# **Mechanical Properties of Polymers**

- i) Stress**
- ii) Strain**
- iii) Molecular basis**

# Hooke's Law

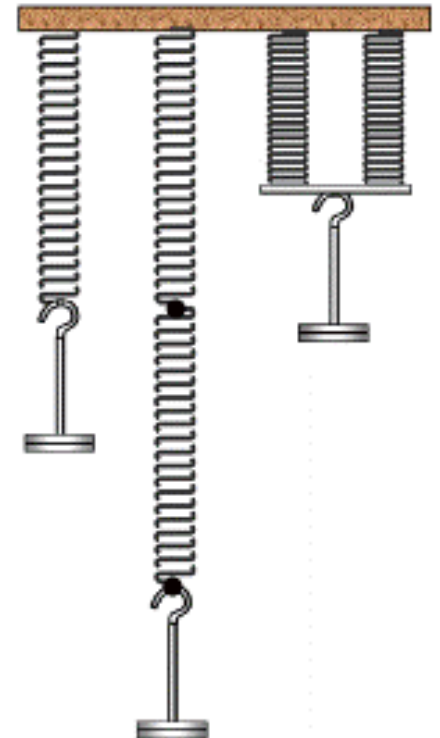
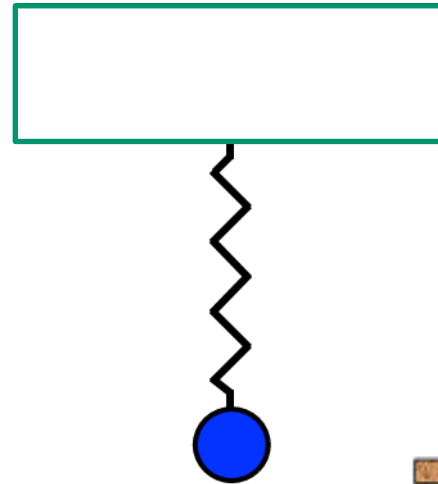
- Hooke's law of springs:

$$F = -k\Delta x \quad E = \frac{1}{2}k\Delta x^2$$

- $\Delta x$  = change in spring length

$$\Delta x = x - x_0$$

- Many materials respond in a linear elastic way under small load
  - when load is taken off, materials “coil” back (elastic change)
  - the “response” and load are linearly related



# Stress

- Forces can be applied on a body in two ways;
- *Body forces* – act on  of a body e.g. gravity, inertia
- *Surface/contact forces* –act on the **surface** of the body but their effect is transmitted to atoms/molecules  e.g. pressure, stress
- For analytical purposes, consider the body as a macro-scale continuum but remember response is at the atomic/molecular level.
- Apply a force,  $F$ , on an area,  $A$ 
  - Stress is force/area

$$\sigma = \frac{F}{A}$$

The units of stress are:

force/area = 1 N/m<sup>2</sup> = 1 Pa

N = mass times acceleration = kg m/s<sup>2</sup>

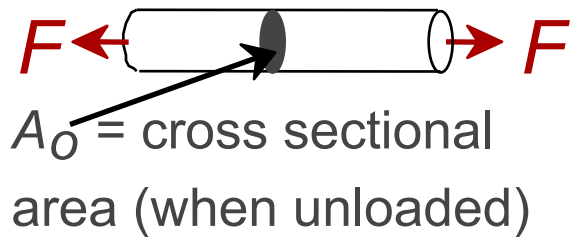
Most often expressed in MPa (10<sup>6</sup> Pa) and GPa (10<sup>9</sup> Pa)

The unit of energy/volume = J/m<sup>3</sup> = (N m)/m<sup>3</sup> = N/m<sup>2</sup> = Pa.

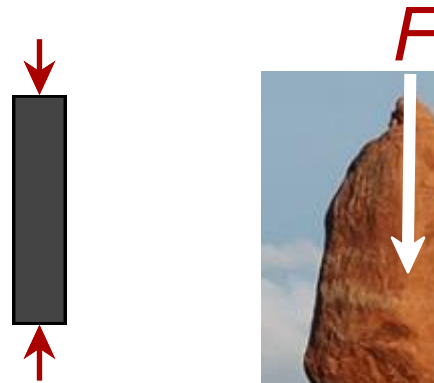


# Normal Stress

- **Tension:**

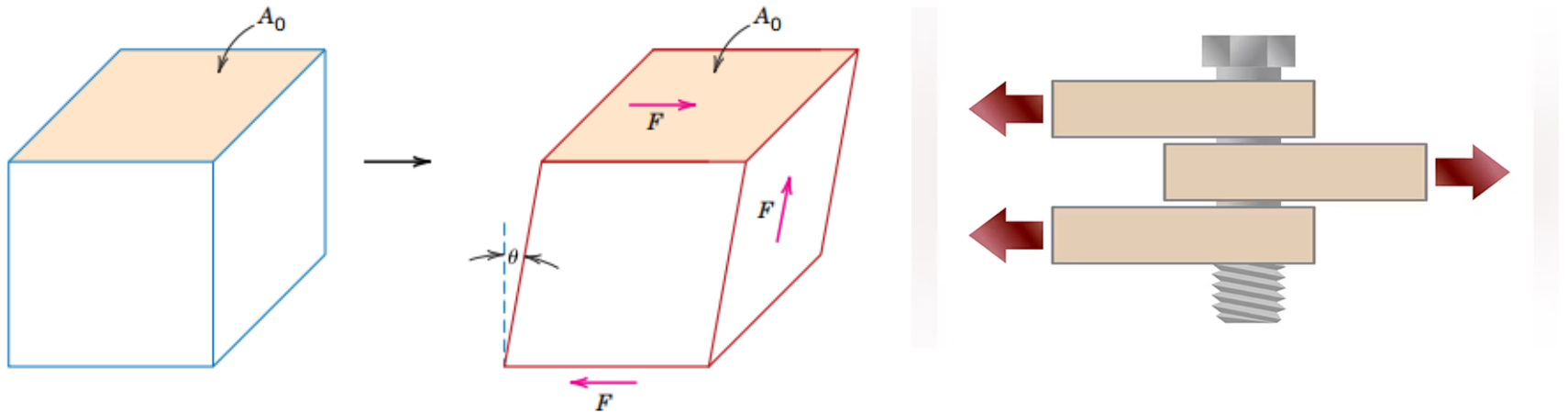


- **Compression:**



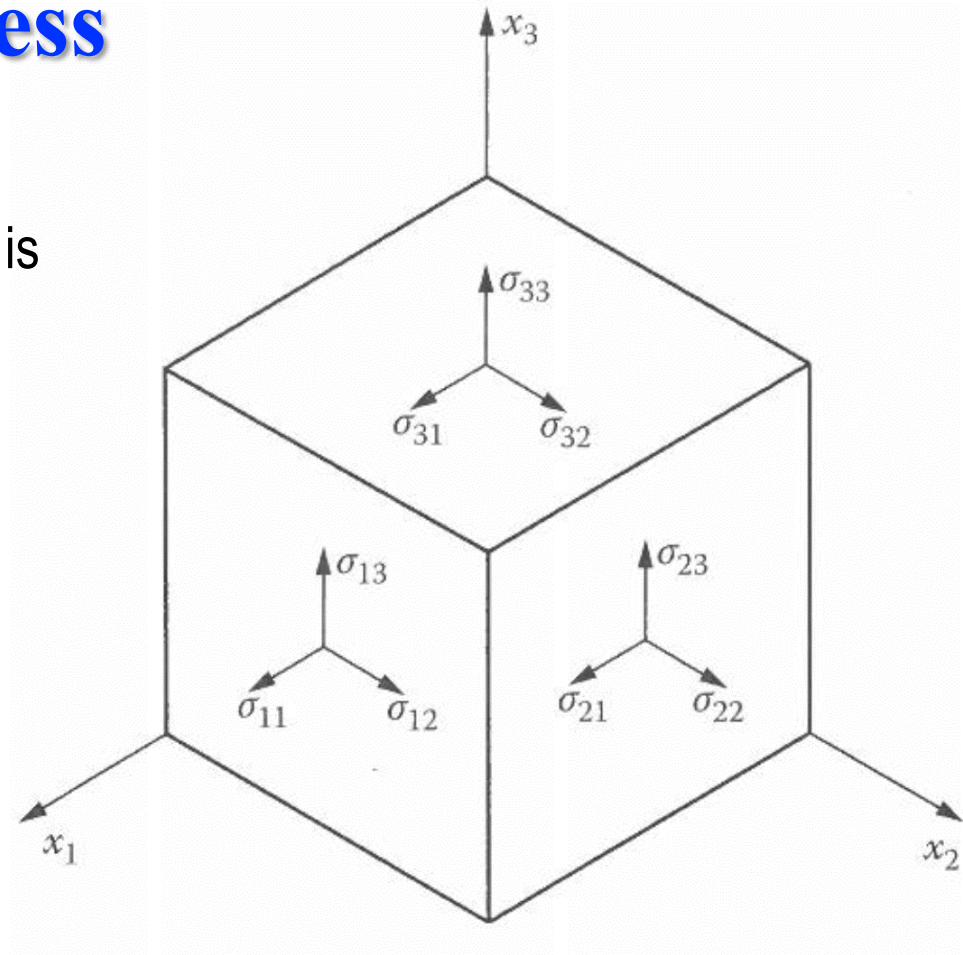
# Shear Stress

- **Shear:**



# Stress

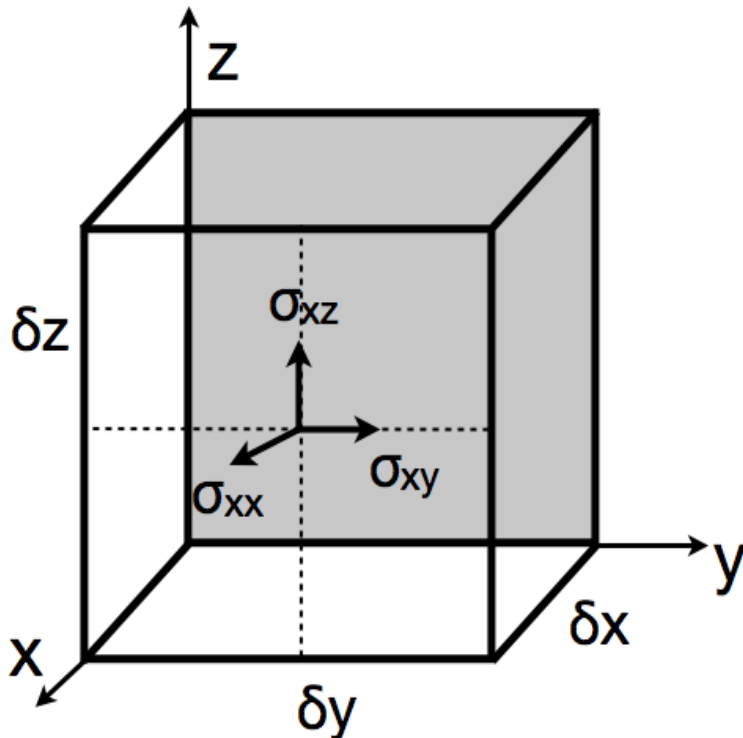
- Consider a body as a macro-scale continuum whose response to force is directional (due to order)
- Let's consider a cube (a simple system)
- Each force is divided by the area on which it acts (recall;  $\sigma = F/A$ ), then resolved into components parallel to the 3 coordinates
- This gives nine components of stress



$$\begin{array}{ccc} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{array}$$

# Stress notation

- The first subscript =  to the plane on which the stress acts
- Second subscript =  of the stress
- $\sigma_{11}, \sigma_{22}, \sigma_{33}$ , are the **normal stresses** i.e.  to the surface they are acting on.  
All **others** are  stresses

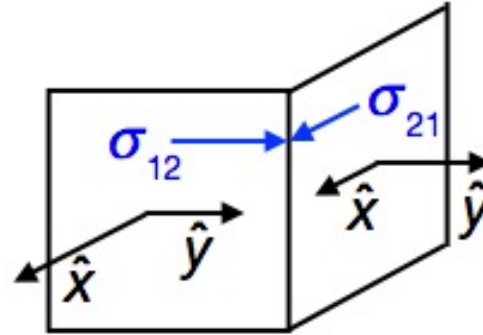


x-axis => 1, y-axis => 2, z-axis => 3

$$\sigma_{ij} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

# But the object is not rotating:

- A body under stress is considered not to be under a *torque* (i.e. not rotating)



- Therefore, the stress tensor must be *symmetric*, i.e.,

$$\sigma_{ij} = \sigma_{ji} \quad \text{i.e. } \sigma_{12} = \sigma_{21}, \sigma_{23} = \sigma_{32}, \sigma_{13} = \sigma_{31}$$

- This reduces the **stress tensor to 6 independent** stress components

$$\sigma_{ij} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} = \begin{bmatrix} \sigma_{11} & \sigma_{21} & \sigma_{31} \\ \sigma_{12} & \sigma_{22} & \sigma_{32} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix}$$

# Stress in polymeric materials

- When considering polymers, stress tensor is divided into
  - Hydrostatic pressure,  $p$
  - Deviatoric components,  $\sigma'_{ij}$
- Hydrostatic pressure (only  stresses, are ) is given by;

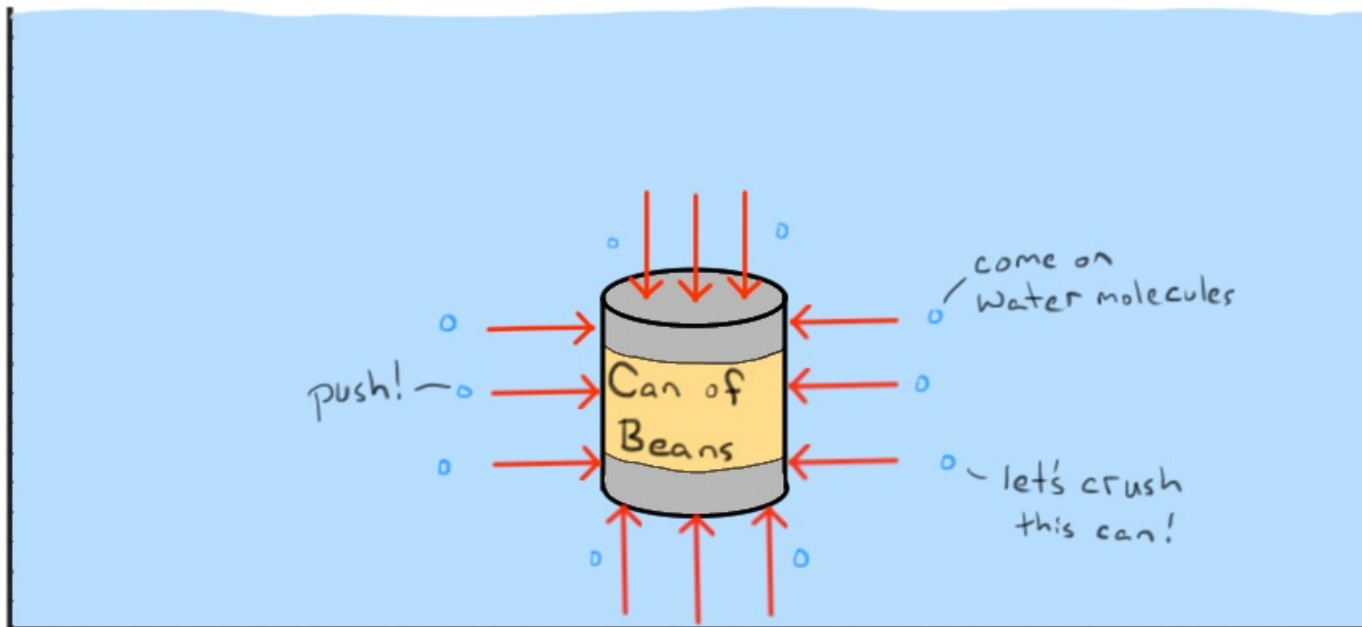
$$p = \frac{1}{3} (\sigma_{11} + \sigma_{22} + \sigma_{33})$$

- Deviatoric (stress-p) stress tensor is given by;

$$\sigma'_{ij} = \begin{bmatrix} (\sigma_{11} - p) & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & (\sigma_{22} - p) & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & (\sigma_{33} - p) \end{bmatrix}$$

$$\sigma' = \sigma - \sigma_{\text{Hydrostat}}$$

# Why "hydrostatic" pressure?



# Strains

- Consider a macroscopic rod of material:



- Initially its length is  $\ell_o = x_2 - x_1$
- Suppose that under load  $x_1 \rightarrow x'_1$   $x_2 \rightarrow x'_2$
- Its new length is  $\ell = x'_2 - x'_1$
- The **strain** is defined as the fractional change in length

$$\epsilon = \frac{\ell - \ell_o}{\ell_o} = \frac{\ell}{\ell_o} - 1 \quad \text{a dimensionless number}$$



# Strain

- When a force is applied to a material, atoms  wrt the force = strain
- Consider a small rod of length,  $l$ , extended by  $\delta l$ . The strain is;

$$e = \frac{\delta l}{l}$$

- The general strain tensor,  $e_{ij}$ , is expressed as a sum of an anti-symmetrical and a symmetrical tensor, i.e.

$$e_{ij} = \frac{1}{2}(e_{ij} - e_{ji}) + \frac{1}{2}(e_{ij} + e_{ji})$$

or simply;

$$e_{ij} = \omega_{ij} + \varepsilon_{ij}$$

where;  $\omega_{ij}$  = rotational strain,  $\varepsilon_{ij}$  = pure strain

- The strain tensor,  $e_{ij}$ , can be extended over 3D – i.e. into  components (recall the stress tensor matrix)

# Strain in 3D

- Given that pure strain is;

$$\varepsilon_{ij} = \frac{1}{2}(e_{ij} + e_{ji})$$

- Then, the matrix can be extended to;

$$\begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{bmatrix} = \begin{bmatrix} e_{11} & \frac{1}{2}(e_{12} + e_{21}) & \frac{1}{2}(e_{13} + e_{31}) \\ \frac{1}{2}(e_{21} + e_{12}) & e_{22} & \frac{1}{2}(e_{23} + e_{32}) \\ \frac{1}{2}(e_{31} + e_{13}) & \frac{1}{2}(e_{32} + e_{23}) & e_{33} \end{bmatrix}$$

- But  $\varepsilon_{12} = \varepsilon_{21}$ ,  $\varepsilon_{23} = \varepsilon_{32}$ , and  $\varepsilon_{13} = \varepsilon_{31}$ , hence like the stress tensor,  $\sigma_{ij}$ , the pure strain tensor,  $\varepsilon_{ij}$ , has only  unique components
- components correspond to **tensile strain**, while  ones are the **shear strain**.

# Stress-strain relations

- For  strains, stress and strain are linearly proportional.
- For  strain, the proportionality constant is,  **$E$** , (Young's modulus)  
$$\sigma = E\varepsilon$$
- For  strain, the proportionality constant is,  **$G$** , (Shear modulus)

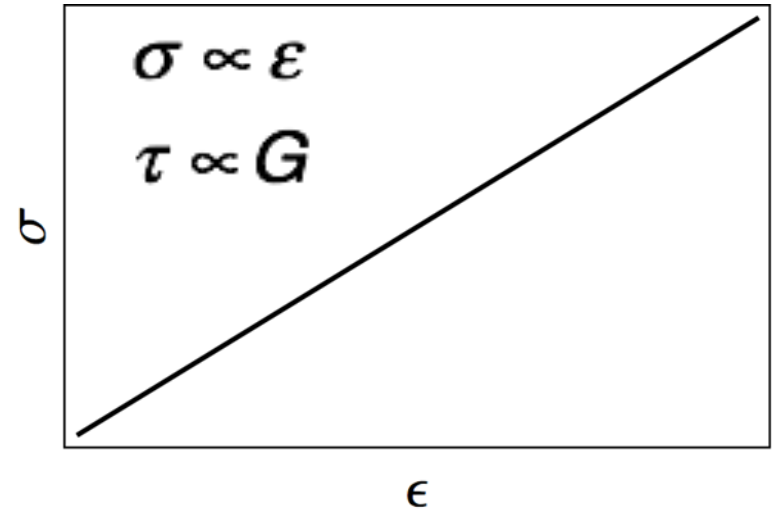
$$\sigma_{ij} = 2G\varepsilon_{ij}$$

- Note:** we used  $\sigma_{ij}$  and  $\varepsilon_{ij}$  to denote stress and strain, but shear strain is sometimes defined as  $\gamma_{ij}$  (i.e. 'Engineering strain') such that;

$$\gamma_{ij} = 2\varepsilon_{ij}$$

hence, shear strain can be expressed as;

$$\tau_{ij} = G\gamma_{ij}$$

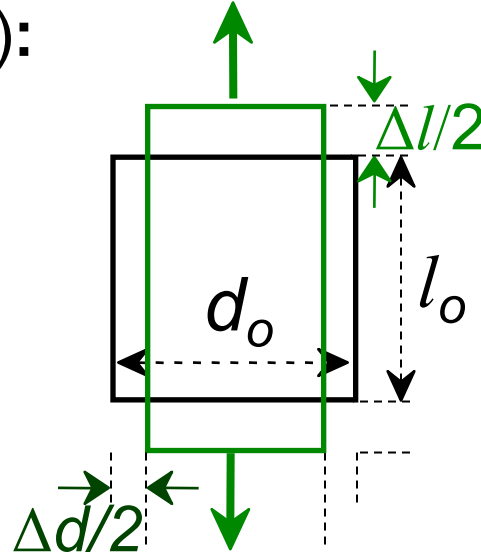


**NB:** I expect you keep track of where strain is abbreviated as  $\gamma_{ij}$  or  $\varepsilon_{ij}$

# Engineering Strain

- **Tensile strain** ( $\varepsilon_z$ ):

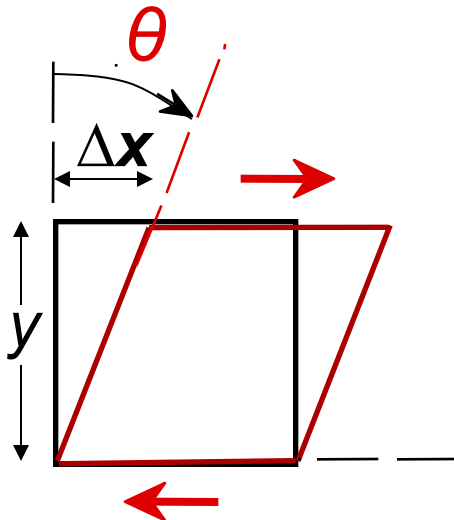
$$\varepsilon_z = \frac{\Delta l}{l_o}$$



- **Lateral strain** ( $\varepsilon_x$ ):

$$\varepsilon_x = \frac{\Delta d}{d_o}$$

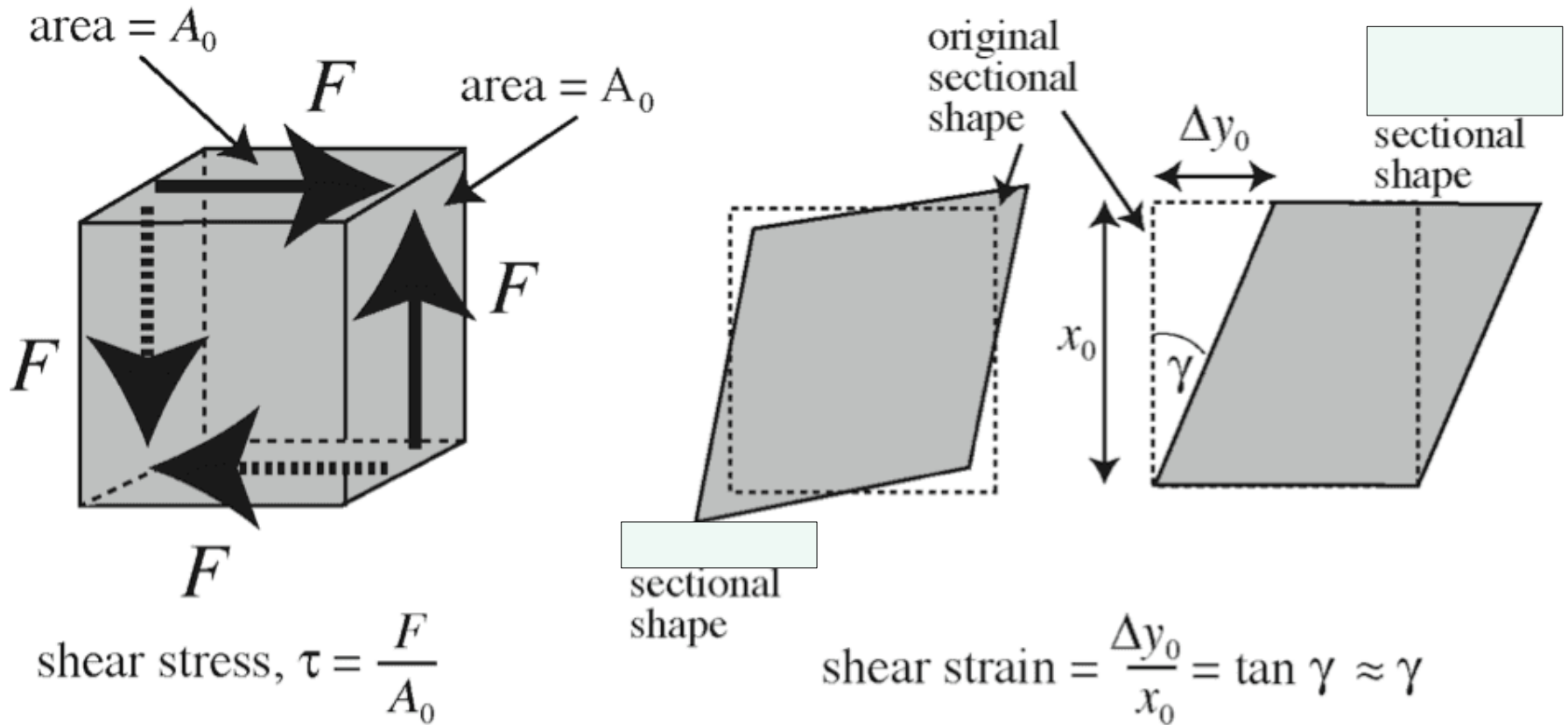
- **Shear strain** ( $\gamma$ ):



$$\gamma = \Delta x / y = \tan \theta$$

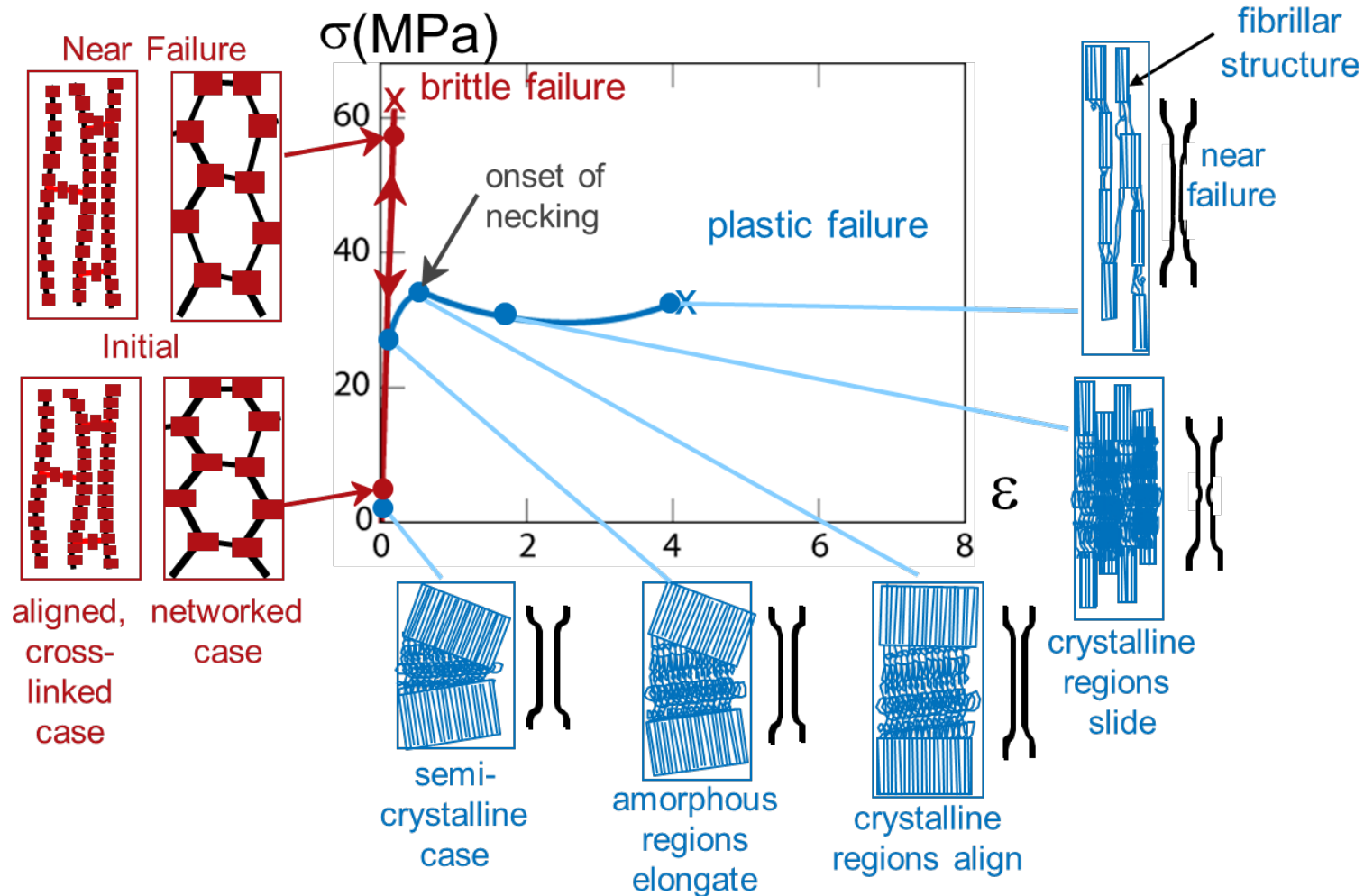
**Both tensile and shear strain are dimensionless**

# Why $\gamma_{ij} = 2\varepsilon_{ij}$ ?



# Stress-Strain Behavior

- How can we relate these to the modulus?



# Bulk constants from Hooke's law

Generalized Hooke's law:

$$\sigma_{11} = A\varepsilon_{11} + B\varepsilon_{22} + C\varepsilon_{33} + D\varepsilon_{12} + \dots$$

$$\sigma_{22} = A'\varepsilon_{11} + B'\varepsilon_{22} + C'\varepsilon_{33} + D'\varepsilon_{12} + \dots$$

A, A', B, B', ... are stiffness constants. Subsequently expressed with contracted notation in tensor form

$$\sigma_{ij} = c_{ij}\varepsilon_{ij}$$

# Bulk constants from Hooke's law

**NB:** Applies only to elastically *isotropic* solids

- Glassy polymers
- Randomly oriented semi-crystalline polymers

Only 2 independent elastic constants:  $c_{11}$  and  $c_{12}$

$$c_{ij} = \begin{pmatrix} c_{11} & c_{12} & c_{13} & c_{14} & c_{15} & c_{16} \\ c_{21} & c_{22} & c_{23} & c_{24} & c_{25} & c_{26} \\ c_{31} & c_{32} & c_{33} & c_{34} & c_{35} & c_{36} \\ c_{41} & c_{42} & c_{43} & c_{44} & c_{45} & c_{46} \\ c_{51} & c_{52} & c_{53} & c_{54} & c_{55} & c_{56} \\ c_{61} & c_{62} & c_{63} & c_{64} & c_{65} & c_{66} \end{pmatrix}$$



# Bulk constants from Hooke's law

**NB:** Applies only to elastically *isotropic* solids

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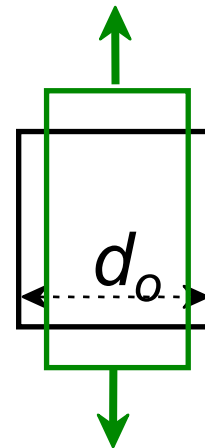
Only 2 independent elastic constants:  $c_{11}$  and  $c_{12}$

$$2c_{44} = c_{11} - c_{12}$$

$c_{12}$  represents the coupling between normal stresses and normal strains in perpendicular directions

$c_{44}$  is easy to interpret:

$$G = c_{44}$$



# Bulk constants from Hooke's law

**NB:** Applies only to elastically *isotropic* solids

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We can express all bulk constants ( $G$ ,  $K$ ,  $E$ ,  $\nu$ )  
in terms of  $c_{44}$  and  $c_{12}$ .

# Bulk constants from Hooke's law

**NB:** Applies only to elastically *isotropic* solids

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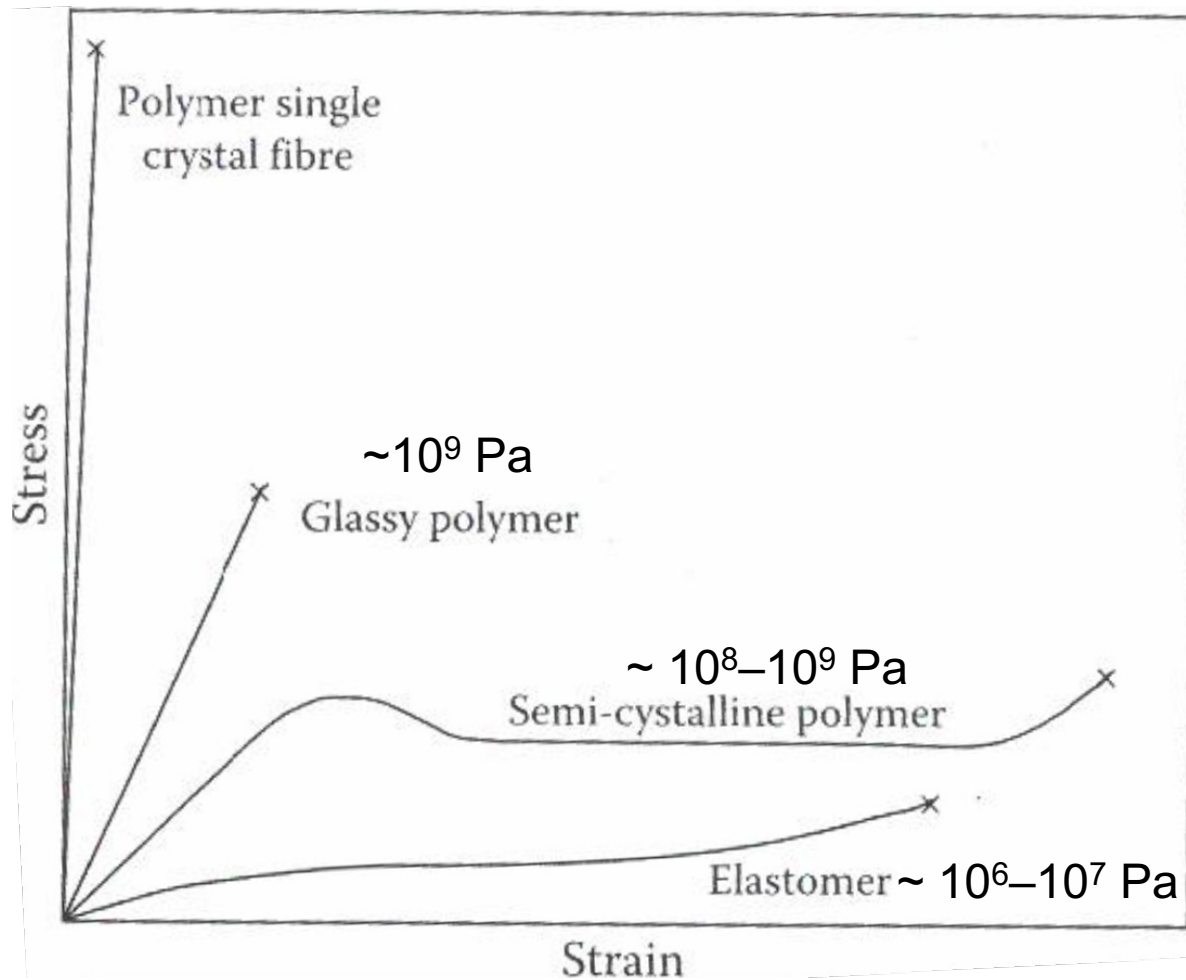
We can express all bulk constants ( $G$ ,  $K$ ,  $E$ ,  $\nu$ ) in terms of  $c_{44}$  and  $c_{12}$ . And therefore relative to one another! For example:

$$G = \frac{E}{2(1 + \nu)}$$

$$K = \frac{E}{3(1 - 2\nu)}$$

# Elastic deformation of Polymers

- $E$  can be determined from the initial slope of these curves
- Polymer fibers are anisotropic, hence, the need to consider directions.  $E_{\parallel} \sim 10^{11}$  and  $E_{\perp} \sim 10^9$
- Comparable to steel! ( $E \sim 10^{11}$  Pa)



So, how do we understand deformations in polymers?

➤ Start with the polymer chains

Why is  $E_{\parallel} > E_{\perp}$  in polymer fibers?

**GROUP ACTIVITY!**